

ASIN FATHIMA S

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CAREER OBJECTIVE

Enthusiastic and detailed oriented Computer Applications Graduate with a strong foundation in Python, AI, Web Development, IoT, and Embedded Systems. Hands-on experience through internships and technical projects, with strong problem-solving, teamwork, and leadership skills.

TECHNICAL SKILLS

Programming Languages: Python | Java | C | C++

Web Technologies: HTML5 | CSS3 | JavaScript

Frameworks & Libraries: React.js (Learning) | Node.js (Learning) | Express.js (Learning)

Full Stack Development: MERN Stack (Learning)

Database: SQL | MySQL

Version Control: Git | GitHub

Cloud & DevOps: AWS (Learning) | Docker (Learning) | Kubernetes (Learning) | DevOps (Learning)

Software Development: Object-Oriented Programming (OOP) | Data Structures & Algorithms |
REST APIs | Software Development Life Cycle (SDLC) | Database Management Systems (DBMS)

IoT & Embedded Systems: NodeMCU (ESP8266) | Arduino IDE | Embedded Systems | IoT Development

Operating Systems: Linux | Windows

Networking: Computer Networks | TCP/IP

Tools: Visual Studio Code | Eclipse IDE | Microsoft Office Suite Programming Languages

EDUCATION

Bachelor of Computer Applications (BCA)

2023– 2026

St. Joseph's College for Women, Tirupur (Bharathiar University)

CORE COMPETENCIES

- IoT & Embedded Systems
- Software Testing & Debugging
- Analytical Thinking
- Team Collaboration
- Communication Skills
- Time Management
- Adaptability & Continuous Learning

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INTERNSHIP & TRAINING

NOBEL SOFTWARE SOLUTIONS, Tirupur | Full Stack Development Trainee

May 2026 – Present

- Currently undergoing an intensive Full Stack Development training program at Noble Software Solutions, Tirupur.
- Gaining hands-on experience in HTML, CSS, JavaScript, React.js, Node.js, Express.js, MySQL, JSON, REST APIs, Git, and modern web application development.
- Learning front-end and back-end development, database management, API integration, web application deployment, and full-stack project development using industry-standard tools and best practices.
- Developing responsive, scalable, and database-driven web applications through real-world projects.

ATS Techno soft, Coimbatore | Course: AI with Python

May 2025 – June 2025

- Completed intensive training in Artificial Intelligence using Python with practical exposure to core AI concepts and programming fundamentals.
- Gained hands-on experience in developing AI-based solutions through structured, project-oriented learning.
- Strengthened logical thinking, analytical skills, and problem-solving abilities by working on real-time use cases and coding exercises.
- Improved understanding of Python programming and its application in building intelligent systems.

PROJECTS

Project Number: 01

Title: MP3 Tag Reader and Editor

JULY 2026 – AUGUST 2026

Project Brief

Developed a command-line MP3 Tag Reader and Editor application to read, display, and modify ID3 metadata from MP3 files. The application supports viewing information such as song title, artist, album, year, track number, and genre, along with editing selected metadata fields using command-line arguments. Implemented binary file handling techniques to read, modify, and preserve MP3 metadata while maintaining the remaining MP3 data.

Technologies Used

*Advanced C Programming | Structures | File Handling | Pointers | String Handling | Bitwise Operations |
Command-Line Arguments | ID3 Metadata | Binary File Handling.*

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Key Challenges & Learnings

- ✓ Implemented ID3 tag validation and metadata extraction from MP3 files using binary file handling techniques.
- ✓ Used structures and pointers to efficiently organize and manage MP3 metadata such as title, artist, album, year, Track number, and genre.
- ✓ Implemented metadata editing by identifying specific ID3 frames and replacing their contents while preserving the remaining MP3 data.
- ✓ Gained practical experience in file pointers, binary data processing, bitwise operations, command-line arguments, and debugging file-based applications.

Project Number: 02

Title: Image Steganography Using LSB Encoding & Decoding

MAY 2026 – JUNE 2026

Project Brief

Built an Image Steganography application that enables secure hiding and retrieval of text messages inside BMP image files using the Least Significant Bit (LSB) encoding technique. Implemented efficient encoding and decoding mechanisms to embed secret data into image files while preserving the integrity and visual quality of the original image. The application accurately extracts the embedded information through the decoding process.

Technologies Used

Advanced C Programming | File Handling | Bitwise Operations | Pointers | Structures | Functions | Command-Line Arguments | BMP File Handling.

Key Challenges & Learnings

- ✓ Implemented the LSB encoding and decoding algorithm to securely embed and extract secret data from BMP image files.
- ✓ Gained a strong understanding of BMP file structure, image headers, and binary file manipulation through practical implementation.
- ✓ Performed bitwise operations while maintaining the integrity and visual quality of the original image.
- ✓ Improved practical knowledge of modular programming, file handling, debugging, and command-line application development in C.

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Project Number: 03

Title: Address Book Management System

MARCH 2026 – APRIL 2026

Project Brief

Built a command-line Address Book Management System for efficient contact management with features such as adding, searching, editing, deleting, and displaying contact information. Implemented file handling techniques to save and load contact records permanently, enabling data retrieval and management across multiple program executions.

Technologies Used

Advanced C Programming | Structures | File Handling | Functions | Pointers | String Handling | Command-Line Interface (CLI).

Key Challenges & Learnings

- ✓ Designed a structured contact management system using C structures to organize and store contact information efficiently.
- ✓ Implemented file handling techniques to save and retrieve contact records, ensuring data persistence between program executions.
- ✓ Developed search, edit, and delete functionalities using string manipulation and logical programming techniques.
- ✓ Gained practical experience in modular programming, input validation, debugging, and developing menu-driven applications in C.

Project Number: 04

Title: Guardian Care Autonomous Safety Network – IoT

DEC 2025 – FEB 2026

Project Brief

Developed an IoT-based security system for automated monitoring of restricted areas. Integrated motion and distance sensors with NodeMCU (ESP8266) to enable real-time detection and immediate alert notifications.

Technologies Used

IoT | Embedded Systems | NodeMCU (ESP8266) | PIR Sensor | HC-SR04 Ultrasonic Sensor | Buzzer | LED | Wi-Fi | Arduino IDE.

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Key Challenges & Learnings

- ✓ Integrated PIR and HC-SR04 ultrasonic sensors for real-time motion and distance detection.
- ✓ Used NodeMCU (ESP8266) for sensor data processing and Wi-Fi connectivity.
- ✓ Implemented buzzer and LED alerts for immediate notification during detected intrusions.
- ✓ Designed an automated safety monitoring system with reliable sensor-based detection and real-time response.

Project Number: 05

Title: ELITE EXHAUST CARE- LPG Leakage and Smoke Detection Safety System

DEC 2025 – FEB 2026

Project Brief

Designed and developed an IoT-based Smart Kitchen Safety Monitoring System to enhance household safety by continuously monitoring hazardous conditions such as LPG gas leakage, fire, smoke, and abnormal temperature levels. The system utilizes embedded sensors to detect potential hazards and immediately alerts users through a buzzer and cloud-based notifications. Real-time sensor data is transmitted via Wi-Fi for remote monitoring, enabling quick response to emergency situations and reducing the risk of accidents.

Technologies Used

ESP3266 | Embedded C | Micro Python | MQ-2 Gas Sensor | Flame Sensor | Temperature Sensor | Buzzer | WIFI, Blynk | Firebase | Arduino IDE.

Key Challenges & Learnings

- ✓ Successfully interfaced multiple safety sensors to monitor gas leakage, fire, and temperature in real time.
- ✓ Implemented Wi-Fi-based IoT connectivity for remote monitoring and instant emergency notifications.
- ✓ Calibrated sensor thresholds to reduce false alarms while ensuring reliable hazard detection.
- ✓ Gained practical experience in embedded firmware development, sensor interfacing, IoT communication, and real time safety monitoring systems.
- ✓ Enhanced debugging and hardware integration skills by combining multiple sensors and alert mechanisms into a single embedded solution.

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Project Number: 06

Title: Laser Beam Security Grid – IoT

OCT 2025 – NOV 2025

Project Brief

Designed and developed an IoT-based Laser Beam Security Grid for real-time intrusion detection and security monitoring. The system uses a laser transmitter and LDR sensor interfaced with NodeMCU (ESP8266) to continuously monitor beam interruptions. When an interruption is detected, the system triggers an instant buzzer alarm and utilizes Wi-Fi connectivity for remote monitoring and future integration with notification platforms. The system provides a cost-effective security solution suitable for homes, offices, and restricted areas.

Technologies Used

IoT | Embedded Systems | NodeMCU (ESP8266) | Arduino IDE | LDR Sensor | Laser Transmitter | Buzzer | Sensor Integration | Hardware Interfacing | Wi-Fi Communication.

Key Challenges & Learnings

- ✓ Successfully integrated a laser transmitter and LDR sensor with NodeMCU (ESP8266) for real-time intrusion detection.
- ✓ Implemented sensor-based monitoring to detect laser beam interruptions and trigger an immediate buzzer alarm.
- ✓ Utilized Wi-Fi connectivity for remote monitoring and potential integration with IoT notification platforms.
- ✓ Gained practical experience in embedded systems, sensor interfacing, hardware integration, IoT communication, and developing cost-effective security monitoring solutions.

Project Number: 07

Title: Automatic Irrigation System Using Rain Sensor and Soil Moisture Sensor

AUG 2025 – SEP 2025

Project Brief

Developed an IoT-based Automatic Irrigation System to automate plant watering based on soil moisture levels and rain conditions. Integrated soil moisture and rain sensors with a NodeMCU microcontroller to continuously monitor environmental conditions. Used a relay module to control the water pump automatically, reducing water wastage and preventing over-irrigation. The system supports smart and efficient irrigation through sensor-based automation.

Technologies Used

NodeMCU | IoT | Soil Moisture Sensor | Rain Sensor | Relay Module | Water Pump | Blynk | Sensor Interfacing.

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Key Challenges & Learnings

- ✓ Successfully integrated soil moisture and rain sensors with NodeMCU to monitor soil and environmental conditions in real time.
- ✓ Implemented sensor-based automation to control the water pump according to soil moisture levels and rain conditions.
- ✓ Used a relay module to automate pump operation, helping reduce water wastage and prevent over-irrigation.
- ✓ Gained practical experience in IoT development, sensor interfacing, microcontroller programming, hardware integration, and smart irrigation automation.

Project Number: 08

Title: CRM Jewel Management System – Salesforce CRM

JUNE 2025 – JULY 2025

Project Brief

Developed a Customer Relationship Management (CRM) application using the Salesforce platform to efficiently manage customer information and sales records. Designed reports and dashboards to provide business insights and improve customer relationship management through organized data handling and reporting.

Technologies Used

Salesforce CRM | Reports & Dashboards | Data Management

Key Challenges & Learnings

- ✓ Developed a CRM application to manage customer and sales information efficiently.
- ✓ Created reports and dashboards for business analysis and data visualization.
- ✓ Improved understanding of Salesforce CRM concepts and customer relationship management.

Project Number: 09

Title: Lease Management System – Salesforce CRM

NOV 2024 – MAY 2025

Project Brief

Developed a Salesforce-based Lease Management System to streamline property leasing operations, including tenant management, lease agreements, payment tracking, and automated communication. Implemented Salesforce automation tools to manage business processes, improve data accuracy, and enhance operational efficiency. Created custom objects for Property, Tenant, Lease, and Payment, along with automated email notifications and workflow processes.

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Technologies Used

Salesforce CRM | Apex | SOQL | Lightning App Builder | Salesforce Flow | Validation Rules | Approval Process | Apex Triggers | Scheduled Apex.

Key Challenges & Learnings

- ✓ Designed custom Salesforce objects for Property, Tenant, Lease, and Payment to efficiently manage property leasing data.
- ✓ Implemented Salesforce Flows, Validation Rules, Approval Processes, Apex Triggers, and Scheduled Apex to automate business processes.
- ✓ Integrated automated email notifications and workflow automation to improve communication and operational efficiency.
- ✓ Gained practical experience in Salesforce CRM development, Apex programming, SOQL, process automation, data validation, and CRM-based application development.

Project Number: 10

Title: E-Commerce API Development

FEB 2024 – MARCH 2024

Project Brief

Developed backend APIs for an e-commerce platform as part of Python Stack Development training. Implemented core API functionalities to support product management, data handling, and backend operations. Designed and tested API endpoints to enable efficient communication between the application and backend services.

Technologies Used

Python | REST API | Backend Development | API Development | JSON | HTTP Methods | Data Handling.

Key Challenges & Learnings

- ✓ Developed backend API endpoints to support core e-commerce functionalities and product management.
- ✓ Implemented efficient data handling and structured API responses using JSON.
- ✓ Gained practical experience in backend development, REST API concepts, HTTP methods, and API testing.
- ✓ Improved understanding of designing and integrating APIs for real-world web applications.

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LANGUAGES

- English
- Tamil
- Hindi (Speaking)
- Urdu (Speaking)

ADDITIONAL ACHIEVEMENTS

- Authored and presented a research paper titled “How APIs Work in Today’s World” at the 2nd International Conference on Recent Trends in Multi-Disciplinary Research and Innovation (ICRTMRI’25), held at A.V.P. College of Arts and Science, Tirupur.
- The paper focused on API functionality, request-response mechanisms, API integration, data exchange, and real-world applications in modern software development. The presentation demonstrated strong research, technical writing, analytical, communication, and technical presentation skills.